Samuel Zarr

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Education

University of Wisconsin-Madison 2023 - Fall 2026

*B.S. Computer Sciences & B.S. Molecular Biology - **GPA 3.9***

- Relevant Coursework: Honors Algorithms, Operating Systems, Biology Foundation Models, Computational Network Biology

Experience

Data & Software Engineer - Oshkosh Corporation (via UW–Madison CS Capstone) | Madison, WI 9/26-12/26

- Built an agentic RAG pipeline using LangGraph and Python to query 1,000+ page technical manuals with verified physical-page citations.
- Engineered a multimodal pipeline (OpenCV, EasyOCR, VLMs) to trace wiring and hydraulic connectivity graphs from large schematic foldouts.
- Designed an authoritative local SQLite store with a FastAPI backend and an optional Databricks Delta Lake / AI Search mirror.
- Developed a React/TypeScript UI that jumps to exact PDF page coordinates and explicitly surfaces missing evidence to eliminate hallucinations.

Student Researcher – Yichiao Zhong Lab | Madison, WI 8/25-9/26

- Studying how transformer-based biology foundation models learn and store cellular dynamics.
- Implemented linear probing and Sparse Autoencoders (SAE) to extract internal information from biology models.
- Performed ablations to investigate how different types of information (genomic sequence, spatial, transcriptome) shapes understanding/reasoning

Research Intern – Gerber lab at Harvard Medical School | Boston, MA 6/23-9/24

- Led the processing and statistical analysis of terabytes of genomic data
- Applied a high-performance computing cluster to test various data processing algorithms: MetaPhlAn4, Kraken2, and mOTUs
- Designed benchmarks to measure output quality, finding a method that recognized 30% more taxa without increasing the false positive rate
- Applied Bayesian network models to identify novel microbe interactions
- Benchmarked ML topic models learning relations within multiomic data

Leadership

Wisconsin AI Safety Initiative | UW Madison 2023-present

- Led curriculum, technical upskilling, and operations for a 130+ member organization with a $30k budget
- Organized the 2026 AIMS Symposium, coordinating panels across industry AI researchers, faculty, and policymakers

Skills

Languages: `C++`, `Python (NumPy, pandas)`, `R`, `JavaScript`, `Java`, `Bash`

Tools: Docker, Slurm, HTCondor, PyTorch, SQL, Git, Claude Code, Codex

Hobbies: I like making software, like procedure botanical flora for Latex. Visit my GitHub.